

L^AT_EX tutorial

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Contents

1	Introduction	2
2	Title of the first section	2
2.1	Subsection of the first section	2
3	Now we will learn about lists	2
4	Playing with images	3
4.1	Wrapped Figure	4
5	Text size niye gutaguti	4
6	Cholo Table banai (onek pera)	5
6.1	Aro complicated table	5
7	Cholo Math kori	5
8	Hyperlinks	6
9	Pseudocode	6
10	Code Snippet	6
11	Citation Examples	6

1 Introduction

This is a new document¹

Some text with *emphasis and nested content*

Some text with *italic and nested content*

Some text with **italic and nested content**

Something about *apples* and *oranges*.

A colored text A strike through line

2 Title of the first section

Text of material in the first section

Double newline for a new paragraph.

Third paragraph.

2.1 Subsection of the first section

Text of material in the subsection.

3 Now we will learn about lists

Lists are of two kinds - ordered and unordered

Ordered List

1. Bus
2. Truck
3. Plane

Unordered List

- ! Apple
- Banana
- Orange

Nested List

- (i) An entry
- (ii) Another One
 - (a) this is at a
 - (b) this is at b
- (iii) Wow! Three entries

Dog: member of the genus *Canis*, which forms part of the wolf-like canids, and is the most widely abundant terrestrial carnivore.

¹with a footnote

4 Playing with images

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Ut purus elit, vestibulum ut, placerat ac, adipiscing vitae, felis. Curabitur dictum gravida mauris. Nam arcu libero, nonummy eget, consectetur id, vulputate a, magna. Donec vehicula augue eu neque. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Mauris ut leo. Cras viverra metus rhoncus sem. Nulla et lectus vestibulum urna fringilla ultrices. Phasellus eu tellus sit amet tortor gravida placerat. Integer sapien est, iaculis in, pretium quis, viverra ac, nunc. Praesent eget sem vel leo ultrices bibendum. Aenean faucibus. Morbi dolor nulla, malesuada eu, pulvinar at, mollis ac, nulla. Curabitur auctor semper nulla. Donec varius orci eget risus. Duis nibh mi, congue eu, accumsan eleifend, sagittis quis, diam. Duis eget orci sit amet orci dignissim rutrum.

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Figure types

- ***h*** ‘Here’ (if possible)
- ***t*** Top of the page
- ***b*** Bottom of the page
- ***f*** A dedicated page only for floats



(a) hapi billu



(b) sad billu

Figure 1: Two states of Billu

In Figure 1b, we can a sad billu and in Figure 1a, we can a hapi billu and .
This is a new line.

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum,

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Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

4.1 Wrapped Figure

Nam dui ligula, fringilla a, euismod sodales, sollicitudin vel, wisi. Morbi auctor lorem non justo. Nam lacus libero, pretium at, lobortis vitae, ultricies et, tellus. Donec aliquet, tortor sed accumsan bibendum, erat ligula aliquet magna, vitae ornare odio metus a mi. Morbi ac orci et nisl hendrerit mollis. Suspendisse ut massa. Cras nec ante. Pellentesque a nulla. Cum sociis natoque penatibus et magnis dis parturient montes, nascetur ridiculus mus. Aliquam tincidunt urna. Nulla ullamcorper vestibulum turpis. Pellentesque cursus luctus mauris.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

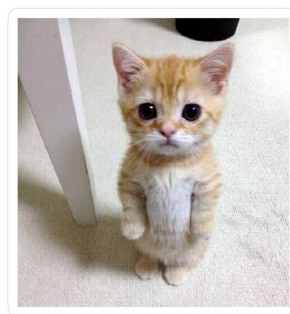


Figure 2: Caption

5 Text size niye gutaguti

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

Ar vallagenaa porashuna korte :(

6 Cholo Table banai (onek pera)

This	is
a	table and it has a really really really really long text

6.1 Aro complicated table

This is a multi column		
1	2	3
A multi row	5	6
	8	9

7 Cholo Math kori

- Let's begin with a formula $e^{i\pi} + 1 = 0$ But we can also do

$$e = \lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = \lim_{n \rightarrow \infty} \frac{n}{\sqrt[n]{n!}}$$

- We can do another:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!}$$

- More Formulas:

$$\vec{v} = \langle v_1, v_2, v_3 \rangle$$

$$\vec{v} \cdot \vec{w}$$

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

- We want to write the solution for the quadratic equation

$$\alpha x^2 + bx + c = 0$$

.

The solution is

$$x = \frac{1}{2\alpha} \times \left(-b + \sqrt{-b^2 \pm 4 \cdot \alpha \cdot c}\right)$$

But there are two solutions actually.

$$\begin{aligned} x_1 &= \frac{1}{2\alpha} \times \left(-b + \sqrt{-b^2 + 4 \cdot \alpha \cdot c}\right) \\ x_2 &= \frac{1}{2\alpha} \times \left(-b - \sqrt{-b^2 - 4 \cdot \alpha \cdot c}\right) \\ &= \frac{-b - \sqrt{-b^2 - 4\alpha c}}{2a} \end{aligned}$$

8 Hyperlinks

Visit [Google](#) for more information

9 Pseudocode

Algorithm 1 Calculate Factorial

Require: $n \geq 0$

Ensure: Returns $n!$

```
1:  $result \leftarrow 1$ 
2: if  $n = 0$  or  $n = 1$  then
3:   return  $result$ 
4: end if
5: while  $n > 1$  do
6:    $result \leftarrow result \times n$ 
7:    $n \leftarrow n - 1$ 
8: end while
9: return  $result$ 
```

10 Code Snippet

```
1 def calculate_fibonacci(n):
2     """Calculate the nth Fibonacci number."""
3     if n <= 0:
4         return 0
5     elif n == 1:
6         return 1
7     else:
8         return calculate_fibonacci(n-1) + calculate_fibonacci(n-2)
9
10 print(calculate_fibonacci(10))
```

Listing 1: A simple Python function

11 Citation Examples

I am citing an example work, [2, 1]

Vaswani et al. [2] introduced transformers in 2017

References

- [1] S. Mirarab, R. Reaz, Md. S. Bayzid, T. Zimmermann, M. S. Swenson, and T. Warnow. Astral: genome-scale coalescent-based species tree estimation. *Bioinformatics*, 30(17):i541–i548, 09 2014. ISSN 1367-4803. doi: 10.1093/bioinformatics/btu462. URL <https://doi.org/10.1093/bioinformatics/btu462>.
- [2] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. Attention is all you need. *Advances in neural information processing systems*, 30, 2017.